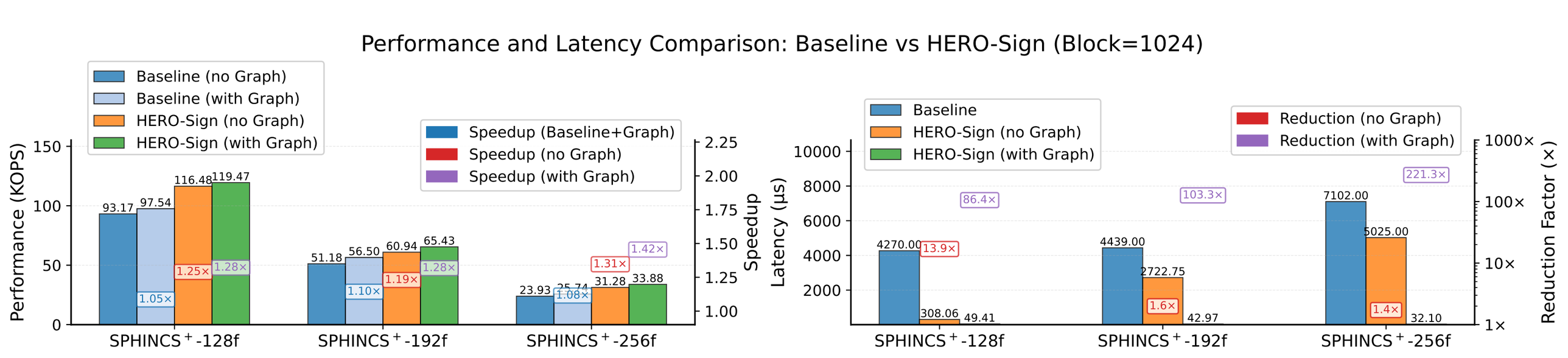
Performance and Latency Comparison: Baseline vs HERO-Sign (Block=1024)
Baseline (no Graph)
Baseline (with Graph)
HERO-Sign (no Graph)
HERO-Sign (with Graph)
Speedup (Baseline+Graph)
Speedup (no Graph)
Speedup (with Graph)
Performance (KOPS)
Speedup
93.17
97.54
116.48
119.47
1.05×
1.25×
1.28×
51.18
56.50
60.94
65.43
1.10×
1.19×
1.28×
23.93
25.74
31.28
33.88
1.08×
1.31×
1.42×
SPHINCS⁺-128f
SPHINCS⁺-192f
SPHINCS⁺-256f
Baseline
HERO-Sign (no Graph)
HERO-Sign (with Graph)
Reduction (no Graph)
Reduction (with Graph)
Latency (µs)
Reduction Factor (×)
4270.00
308.06
49.41
13.9×
86.4×
4439.00
2722.75
42.97
1.6×
103.3×
7102.00
5025.00
32.10
1.4×
221.3×
SPHINCS⁺-128f
SPHINCS⁺-192f
SPHINCS⁺-256f